

Pioneers in Computer Graphics

SIGGRAPH 2022 Retrospective Panel

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ABSTRACT

Join this panel of women pioneers in Computer Graphics to hear about industry and academic view points and career journeys that have contributed to the historic advancements in this field. Learn what opportunities are on the horizon for the women in technology community and how the past can influence the future in this exciting time of growth.

KEYWORDS

computer graphics history, women in computer graphics

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1 OVERVIEW

Five accomplished members across several sectors of Computer Graphics discuss their careers and what lies ahead for their field. Tune in to hear these pioneers informally relay stories from the past, reflect on the nature of success and look to the future. These women pioneers also touch on the challenges of being different, and what it took to get where they are.

Loosely, the panel follows this set of questions:

- (1) Out of the many things you have done, which do you find the most meaningful to yourself and the world?
- (2) Please tell us a story that brings us back in time, to the way things were in your field some time ago, whether from the technology perspective, workplace perspective or other.
- (3) Can you describe the biggest change in your industry over your career, and its new greatest challenge?
- (4) What inspired you to join this field? Why should young people today consider a career in computer graphics and its many industries?

2 BIOGRAPHIES

2.1 Carolina Cruz-Neira

Carolina Cruz-Neira is a pioneer in virtual reality (VR), and also an engineer, researcher, designer and educator. She invented the CAVE immersive virtual environment that has become the standard standard for rear projection-based Virtual Reality systems. She has also advanced the field of VR as a professor at many universities, including Iowa State University, University of Louisiana at Lafayette, University of Arkansas at Little Rock, and she is currently an Agere Chair Professor at University of Central Florida. Carolina has shaped the direction of VR in many ways, including through her research, advocacy, and her many students. Carolina is a member of the National Academy of engineering.

2.2 Holly Rushmeier

Holly Rushmeier is the John C. Malone Professor of Computer Science at Yale University, and has led a distinguished career in computer graphics research, receiving among many other honors the ACM SIGGRAPH Computer Graphics Achievement Award for her work on global illumination, material capture, and the display on high dynamic range images. Her work has enriched our quantitative understanding of every aspect of rendering, from the

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microscopic characteristics of scattering to the broader representation of appearance to the perception of images by the human visual system. Holly has also applied her work on capture to preservation of world's cultural heritage.

2.3 Joan Collins

Joan Collins is a producer and visual effects supervisor, with 40 years of experience in visual effects. She has worked on 26 feature films, managing massive teams and budgets, and ensuring, in her own words, that the movies were getting done. Joan has catalyzed introduction of Computer Graphics technology to the areas it previously did not belong, such as visualization for automotive and airplane designs. Joan was the first woman to received the prestigious Tesla award at the recent Satellite Awards for visionary achievement in filmmaking technology.

2.4 Theresa-Marie Rhyne

Theresa-Marie Rhyne is currently consulting on the practice of applying color theory to digital media and visualization, and is a color maven and digital media guru. In 2015, her book on these subjects was published by CRC Press. In her career, she founded not one,

but two visualization centers: (1) the United States Environmental Protection Agency's Scientific Visualization Center in the 1990s and (2) the Center for Visualization and Analytics at North Carolina State University in the 2000s. She currently writes articles for the *Nightingale*, the journal of the Data Visualization Society.

2.5 Kristine Middlemiss

Kristine has an Honours Bachelor of Computing Degree with a major in Computer Science and a minor in Fine Art from the University of Guelph as well as a certificate in Maya Computer Animation from Humber College. Kristine has previously worked as a Lead Software Developer, Lead Developer Consultant and Senior Technical Director at such companies as IBM/Lenovo, Alias/Autodesk and Rockstar Games. Most recently she has moved back from the LA area where she was working as a Lead Motion Capture and Advanced Technology Engineer at DreamWorks Animation/NBCUniversal, where she received Technical Achievement Awards for Performance Animation Pipeline in 2015, AR/VR Production Asset Transfer Pipeline in 2017 and Mocap-to-Layout Data Delivery Pipeline in 2018. Her most current and exciting role is Chief Technology Officer at Monsters Aliens Robots Zombies, a VFX Studio.